Technology Stack

Date	23 October 2026
Team ID	0827CY231029
Project Name	Waypoint - AI Travel Itinerary Planner
Maximum Marks	2 Marks

Define Your Technology Stack

The selected stack supports a responsive browser interface, typed API requests, optional AI generation and local or hosted data storage.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	JavaScript, React 18.3, Vite 5, React Router 6, CSS; Framer Motion and Lucide.	Reusable pages and components, client routing, animation and icons.
2	Backend / Server-Side	Python, FastAPI, Pydantic, Uvicorn; Gunicorn for hosted startup.	API routes, input validation, OpenAPI documentation and ASGI serving.
3	Database / Data Storage	Supabase Postgres or SQLite; repository adapters.	Relational users, preferences, trips, days and activities; SQLite enables local operation.
4	Cloud / Hosting / Deployment	Render for the API; Vercel for the frontend; Supabase for hosted storage.	Deployment configuration and guides are supplied. Live service URLs are not recorded.
5	Version Control & CI/CD	Git and GitHub; render.yaml and frontend/vercel.json.	Repository history and deploy configuration. Automated CI tests are not supplied.
6	Third-Party APIs / Other Tools	Groq, OpenAI, Hugging Face, optional Ollama; httpx; python-jose and bcrypt.	Generate structured itineraries, call provider endpoints and authenticate users.

AI Selection and Fallback

In automatic mode, available_provider() chooses Groq if configured, otherwise OpenAI, otherwise Hugging Face. Ollama can be selected explicitly. A failed or absent selected model falls back to local rules; the code does not automatically retry every provider after a failure.

Project sources: frontend/package.json; backend/requirements.txt; backend/app/config.py and services/llm/__init__.py; deployment files.